

DATA VISUALIZATION · SUMMER 2026 · FINAL INDIVIDUAL PROJECT

When the World's Shipping Lanes Narrowed

A data-visualization study of global supply chain performance, 2019–2026 — built on IMF PortWatch's satellite AIS record of ~90,000 vessels.

10 analytical questions

Plotly · publication-ready

Interactive Streamlit dashboard

THE STORY IN ONE SENTENCE

Trade didn't shrink — it moved, and not evenly

Between late 2023 and 2024, two of the world's four critical maritime chokepoints constricted at the same time: the Red Sea to conflict, the Panama Canal to drought. The satellite record shows exactly how the fleet absorbed it — and who paid the price.

**-62% /
-76%**

Suez & Bab el-Mandeb transit calls
fell from baseline — still down
roughly half a year+ later

+104%

Cape of Good Hope traffic vs. its
own baseline — the reroute
overshot, it didn't just offset

6.5 months

Panama and the Red Sea/Cape
corridor were constrained
simultaneously — no easy
alternative

~30%

Share of global container calls held
by the top 20 ports — flat since
2019, not rising

The next ten slides derive each of these numbers from the data — one analytical question at a time, each with its own publication-ready figure and an honestly reported finding, including two null results and one reversed hypothesis.

DATASET

IMF PortWatch — real-world, rich, and varied

Daily port-call and trade-volume estimates for 2,065 ports and daily transit counts for 28 major chokepoints, derived from satellite AIS signals broadcast by ~90,000 vessels. Published by the IMF, updated weekly.

Type	Columns
Temporal	date, year, month, day — daily, 2019 → present
Spatial	port coordinates, country, ISO3, 28 named chokepoints
Categorical	5 cargo classes; port systemic-importance proxy
Numerical	port calls, import/export volume, transit capacity

Secondary sources joined in the analysis

- **World Bank LPI** — logistics quality per country (Q7)
- **NY Fed GSCPI** — monthly global supply-chain pressure index (Q10)

Caveat, stated up front: port calls are observed vessel movements; trade volumes are *model-based estimates* from vessel type/capacity/draft, not customs records. Every finding here is framed as relative change over time for exactly this reason.

METHODOLOGY

One shared toolkit, used consistently across all ten questions

Analytical techniques

- **Baseline indexing** — rebase each series to its own pre-crisis average = 100, so passages of very different size are visually comparable
- **Centred rolling smoothing** — removes the weekly port-operations cycle without shifting turning points
- **Coefficient of variation** — ranks volatility across chokepoints of any size
- **Recovery half-life** — days to climb back to half the post-shock gap
- **HHI / top-20 share** and **lead-lag cross-correlation** for concentration and the GSCPI comparison

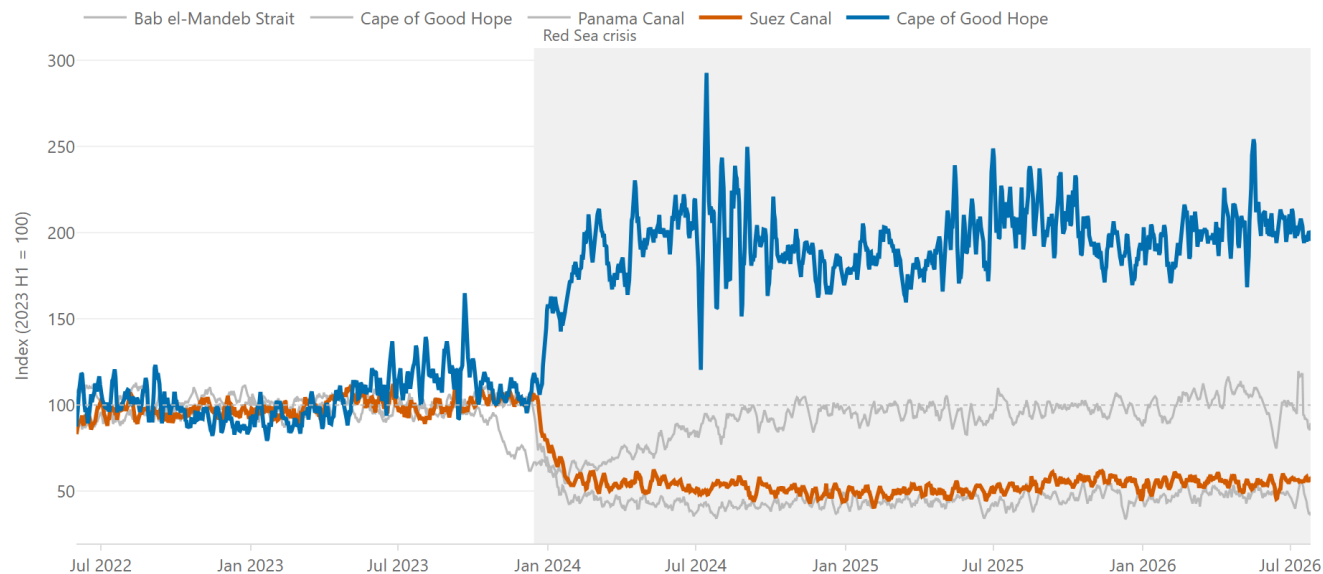
Design system (course requirements)

- Okabe–Ito **CVD-safe palette** — one shared Plotly template for every figure
- **Muted grey for context**, one highlight colour per chart
- No gridlines, no chart junk, clean white background
- **Titles state the takeaway** — not the axes
- Findings reported honestly — including two null results and one reversed hypothesis

Did traffic lost at Suez reappear at the Cape of Good Hope?

Suez traffic collapsed - and reappeared around the Cape of Good Hope

Daily transit calls, 7-day mean, indexed to each chokepoint's Jan-Jun 2023 average = 100

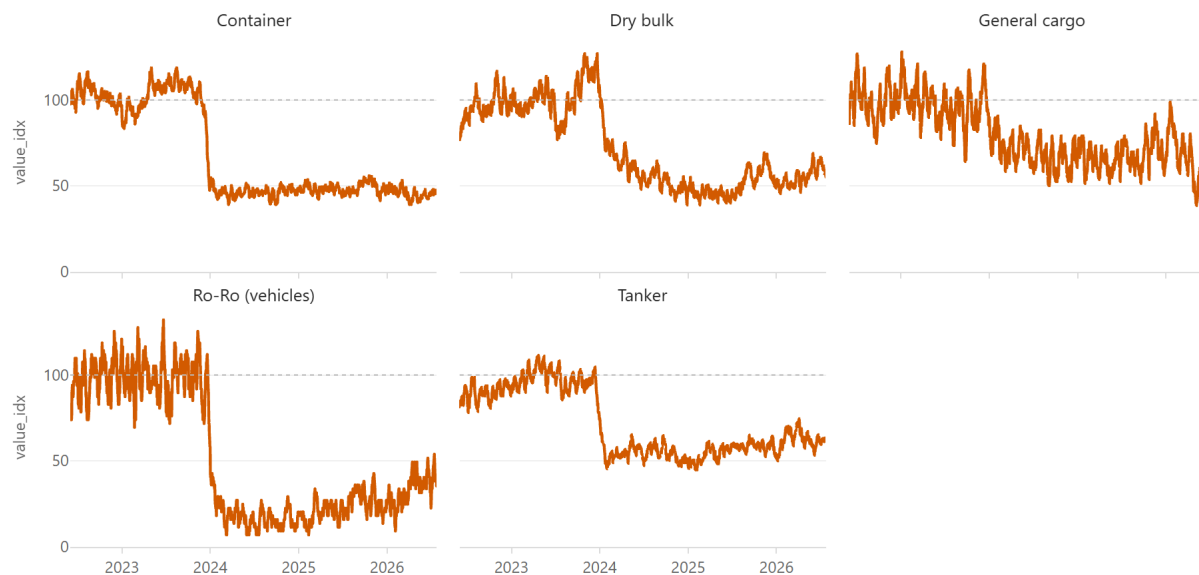


What we found. Suez and Bab el-Mandeb fell 62% and 76% from baseline and are still running roughly half their pre-crisis level. The Cape didn't just absorb the loss — it overshoot it, running at **2×** its own baseline. The offset is not proportional: more Cape transits are needed to move the same cargo one Suez transit used to carry.

Did every cargo type reroute the same way?

Vehicle carriers abandoned the Red Sea hardest; general cargo held on longest

Suez Canal transit calls by vessel class, 14-day mean, indexed to Jan-Jun 2023 = 100

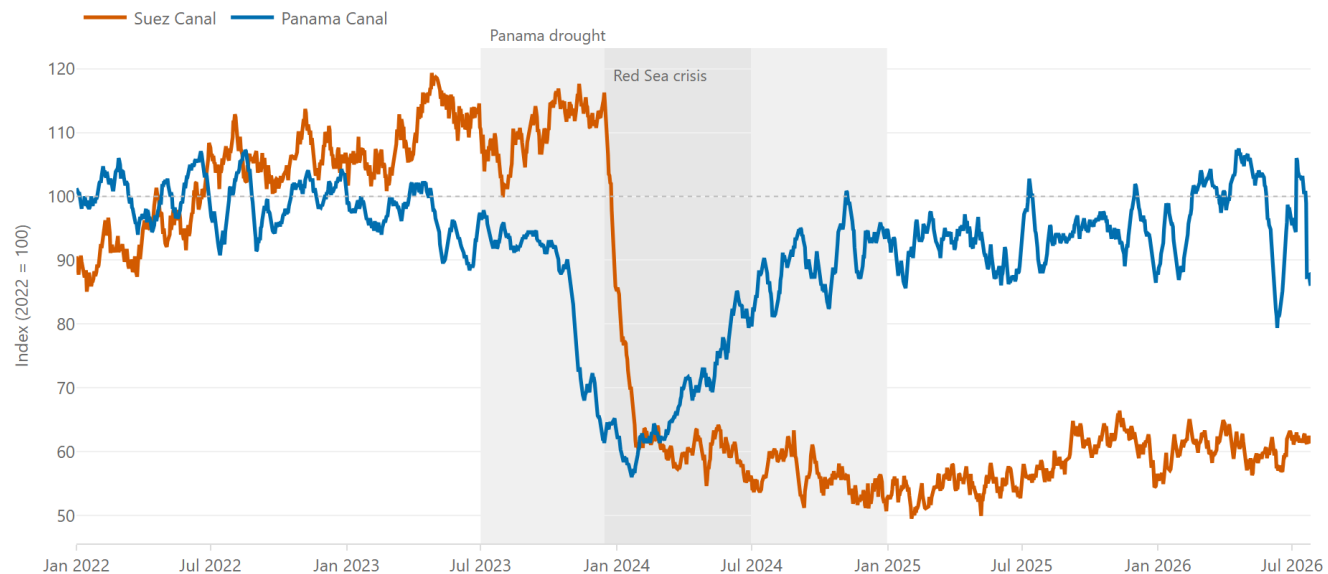


What we found — the hypothesis didn't survive contact with the data. We expected container traffic to flee first. Instead, **Ro-Ro (vehicle carrier) traffic fell hardest, down 93%**, while general cargo fell least (–40%). Plausibly, Ro-Ro cargo is both schedule-sensitive and the vessels comparatively exposed to the reported attacks — the strongest incentive to reroute entirely rather than accept the risk.

Did Panama and the Red Sea compound each other?

Both Asia-to-US-East-Coast routes narrowed at once, leaving no easy alternative

Transit calls, 14-day mean, indexed to each chokepoint's 2022 average = 100

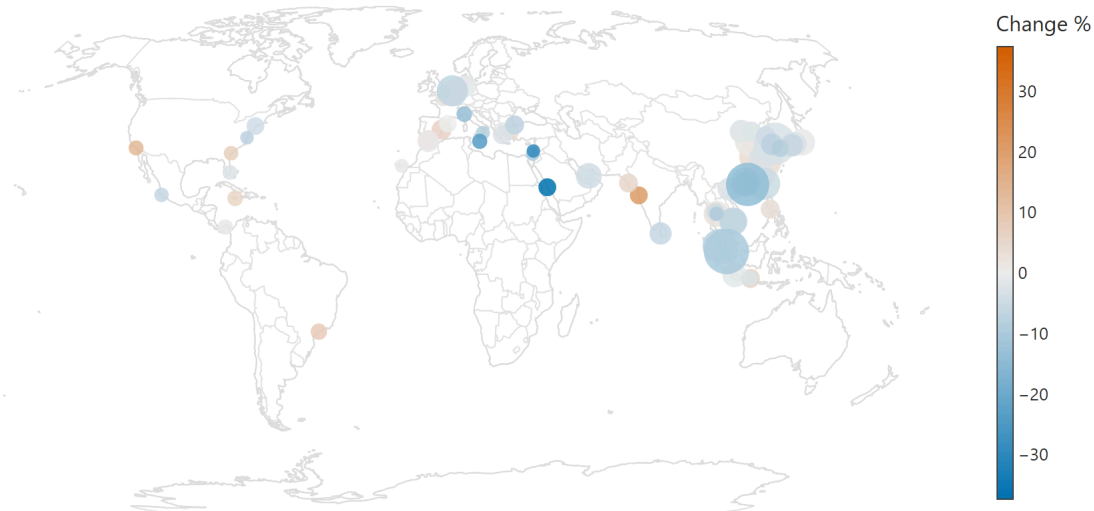


What we found. The two disruption windows overlapped for **6.5 months** (15 Dec 2023 – 30 Jun 2024). Panama transits fell 53% from their pre-drought baseline for the entire overlap, at the same time Suez/Bab el-Mandeb were down 62–76%. Both of the fastest Asia–US–East-Coast routes were constrained at once — the crux of the whole story.

Which ports won and lost — and where?

The winners and losers cluster along the re-routed corridors

Change in weekly container calls, 2024 vs 2023 - bubble size = baseline volume

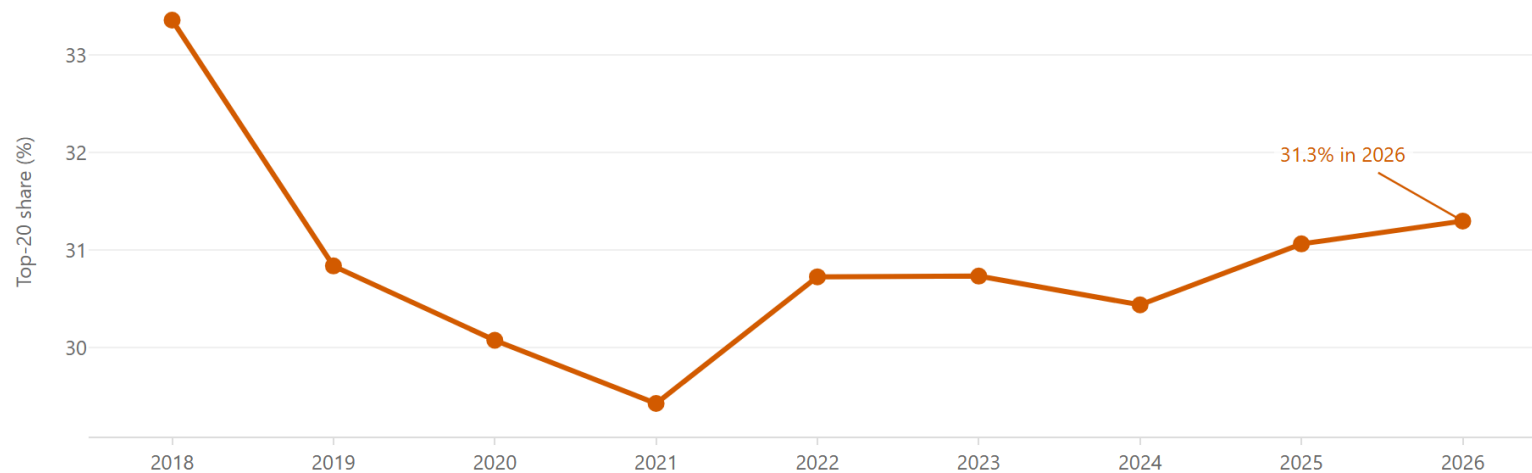


What we found. The losers are geographically coherent and point straight at the Red Sea: **King Abdullah Port (-81%)**, **Aqaba (-71%)**, and **Jeddah (-37%)** — all Red Sea-facing gateways that lost transiting feeder traffic. Winners cluster on the Mediterranean/Black Sea side (Iskenderun +76%, Mersin +36%, El Sokhna +42%), though not every gainer shares the same cause.

Is global container traffic becoming more concentrated?

The 20 busiest ports handle a persistently large share of global container traffic

Share of all container port calls handled by the top 20 ports, by year



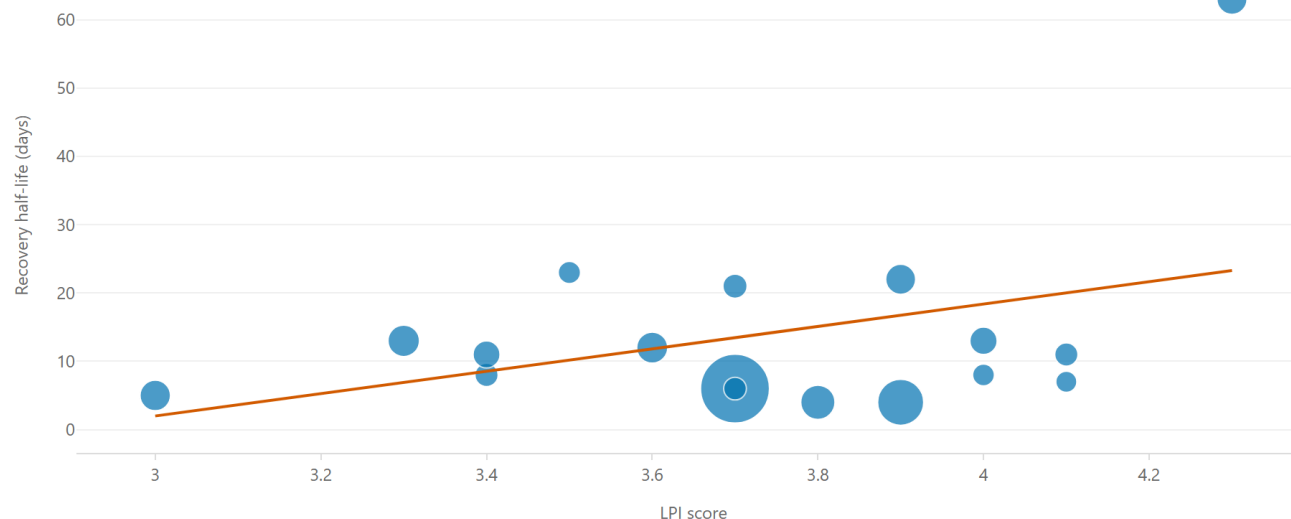
Source: IMF PortWatch

An honest null result. The top-20 ports' share of global container calls is essentially flat — 30.9% in 2019 vs. 31.3% in 2026, drifting inside a narrow 29–31% band even through the crisis. Concentration didn't rise because of the disruption. But a stable ~30% share sitting in just 20 of ~2,065 ports is itself the finding: that structural exposure never went away.

Does logistics quality buy resilience?

Logistics quality only weakly predicts recovery speed - Singapore is the clear outlier

World Bank LPI score vs days to recover half the post-shock gap in container calls

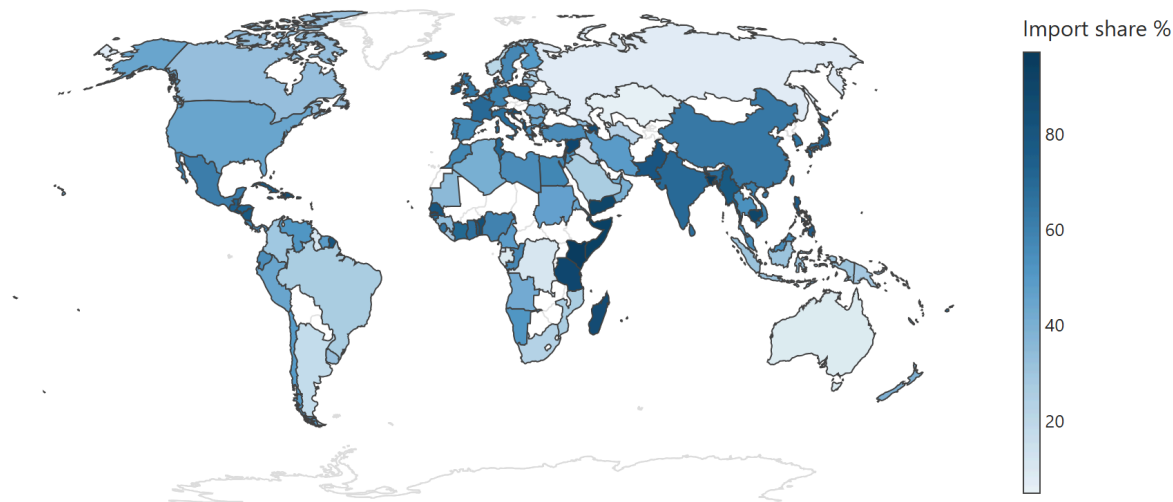


What we found. A weak-to-moderate positive correlation ($r = 0.39$, $n = 17$) in the predicted direction — but **Singapore**, the highest-LPI country in the sample, had the single **longest** recovery (63 days), plausibly because it's itself a Suez/Bab el-Mandeb transshipment hub. Suggestive at best, not a demonstrated causal link, with one shock and a small sample.

Which economies are most exposed to a single chokepoint?

Import-dependent economies carry the most chokepoint risk

Seaborne imports as a share of total seaborne trade, full record

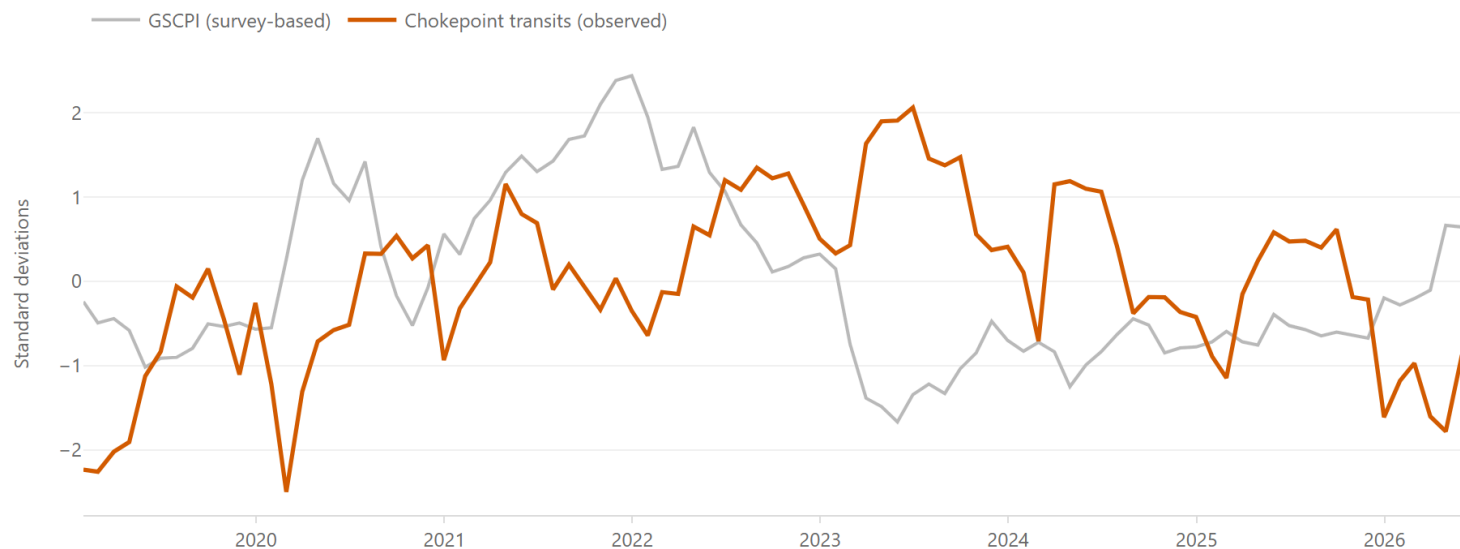


What we found. The proxy cleanly splits two economy types: small island/coastal economies — French Polynesia, Cayman Islands, Haiti, Puerto Rico (93–97% import-dependent) — versus resource exporters — Kazakhstan, Russia, Australia (5–9%). A trade-balance proxy, not a measured routing exposure, but a clear structural pattern regardless.

Does the Fed's supply-chain index track the ships?

The survey-based pressure index and the observed fleet tell related but not identical stories

Both series standardised (z-scores), monthly



Sources: IMF PortWatch; Federal Reserve Bank of New York

What we found. Best-fit lag of **+6 months**, but only weakly correlated ($r = 0.25$). GSCPI barely reacts to the chokepoint-level disruption that the AIS data shows clearly around 2023–24 — the survey-based index and the physically observed fleet tell related but distinct stories, not strongly enough to claim either reliably leads the other.

INTERACTIVE DASHBOARD

A curated, five-tab Streamlit dashboard

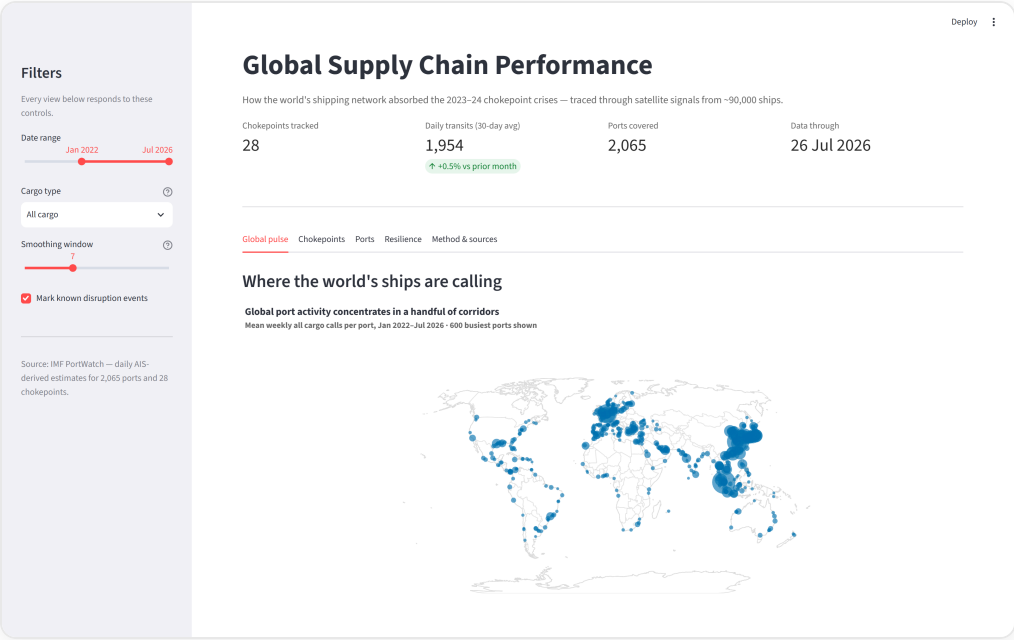
Global pulse · Chokepoints · Ports · Resilience · Method & sources — every view responds live to a shared set of filters: date range, cargo type, smoothing window, and disruption event markers.

Same CVD-safe palette as the notebook

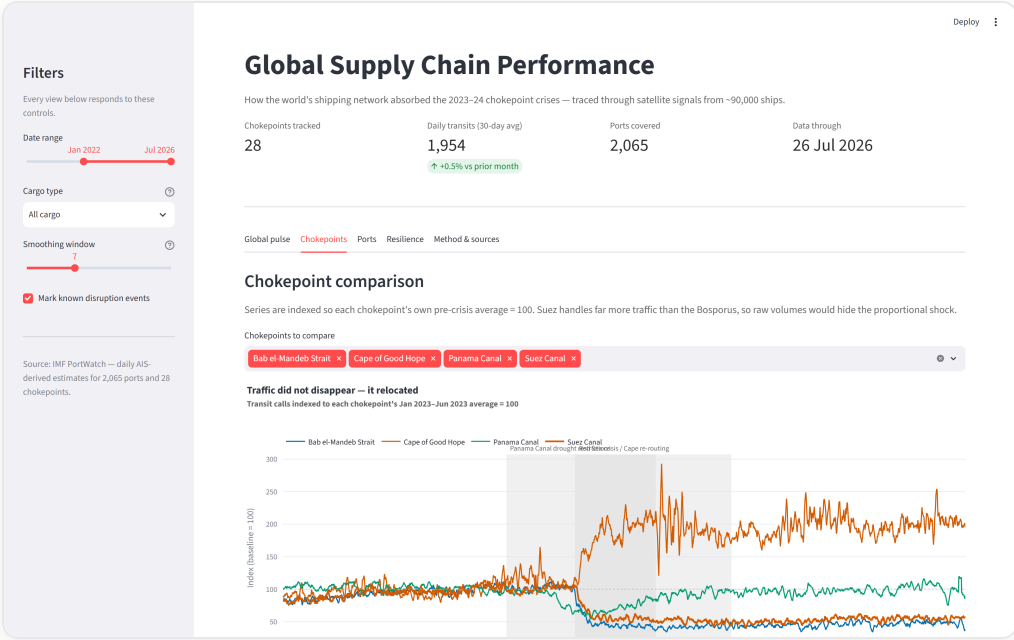
Live filtering, not static charts

Deployed on Streamlit Community Cloud

Global pulse & chokepoint comparison

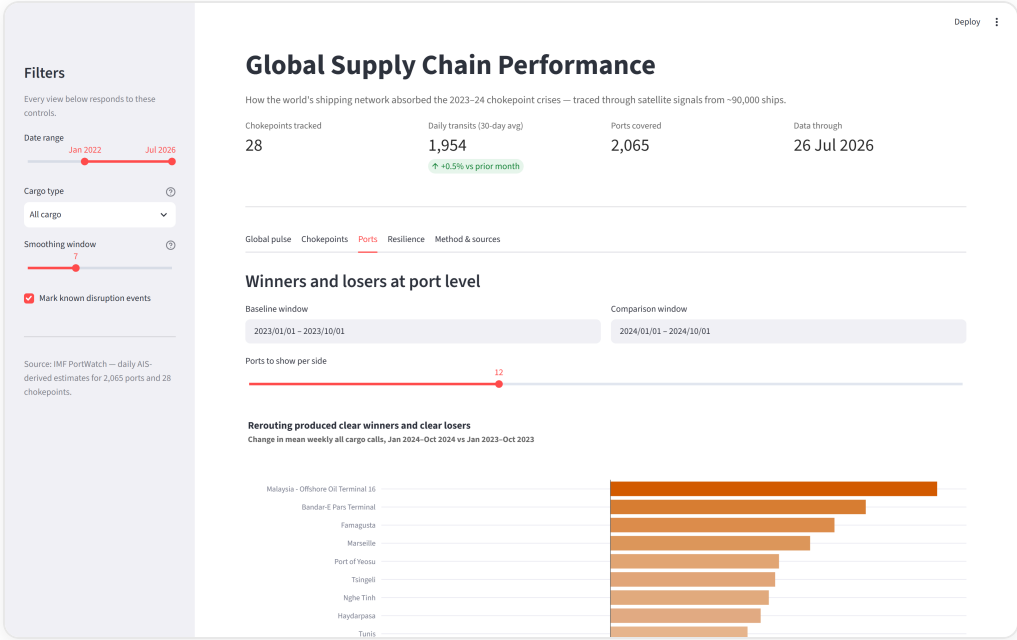


Tab 1 — where the world's ships are calling, plus aggregate transit volume

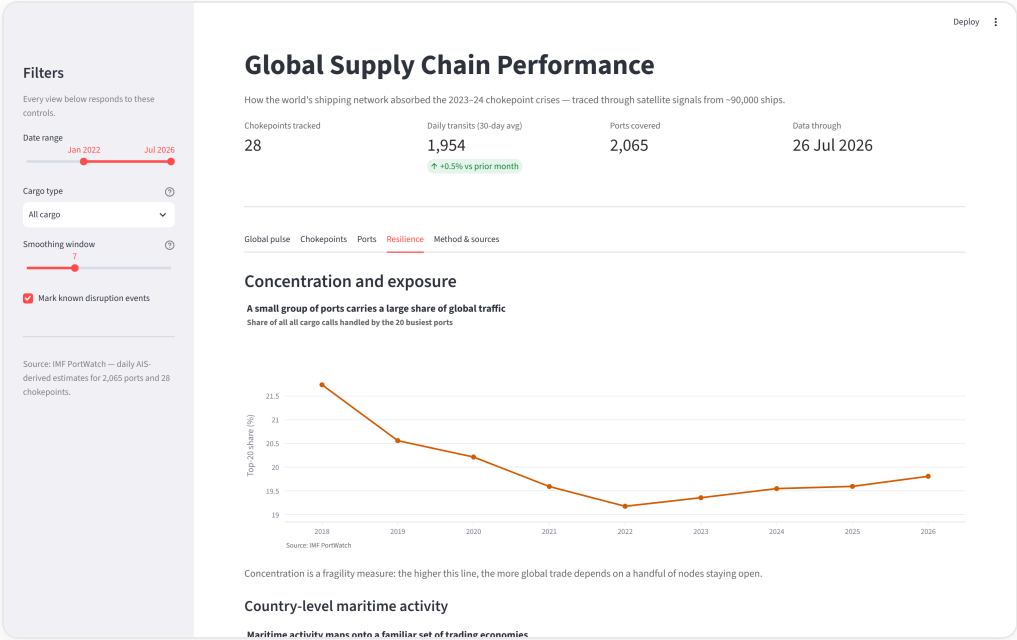


Tab 2 — user-selected chokepoints, indexed and compared live

Port-level winners & losers, on demand



Tab 3 — pick any baseline/comparison window, see the ports that moved



Tab 4 — concentration trend and country-level exposure

CONCLUSIONS

One story, told from five directions

- 1 The shock was real and large** — Suez/Bab el-Mandeb fell 62–76% and stayed there.
- 2 It was a rerouting, not a contraction** — the Cape overshoot its baseline; cargo types responded unevenly, overturning our initial hypothesis.
- 3 It was unusually hard to escape** — Panama and the Red Sea narrowed together for 6.5 months, removing the two fastest alternatives at once.
- 4 Costs landed unevenly, along clear geographic lines** — while structural concentration in the top-20 ports stayed flat throughout.
- 5 Resilience is harder to pin down than the shock itself** — logistics quality, port rhythm, and the survey-based pressure index all produced honest, modest, or null results rather than clean answers.

LIMITATIONS

What this analysis can't tell you

- PortWatch volumes are **AIS-derived estimates**, not customs records — every finding is framed as relative change for this reason.
- **Vessel counts are not capacity** — a shift to fewer, larger ships would understate volume.
- Event dates shown as shaded bands are drawn from **public reporting**, used as context, not as data.
- Q7's correlation is drawn from a **single shared shock and n = 17** — suggestive, not causal.
- Q8's exposure measure and Q9's port classification are **documented modelling choices**, not measured routing assignments.

THANK YOU

One dataset. Ten questions. One story.

 Analysis notebook & source — github.com/bopanna012/Global-Supply-Chain-Performance-Dashboard

 Live dashboard — global-supply-chain-performance-dashboard-7en4lf4cnljnwckrwqeg.streamlit.app

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